

Problem Set 4: Enzyme Kinetics

Due: Sunday, October 18 at 11:59 PM

Value: 25 points

Format: Five-question problem set

Purpose

Connect collision frequency, active-site chemistry, saturation, inhibition, temperature, and pH to measured reaction rates.

Your task

- Q1 (5): Explain why increasing substrate eventually stops increasing rate even though substrate molecules remain available.
- Q2 (5): Estimate V_{max} from the control data and identify the lowest substrate concentration that produces at least half of V_{max} .
- Q3 (5): Compare the inhibitor data with control and decide whether the pattern is more consistent with competitive or noncompetitive inhibition.
- Q4 (5): Predict how enzyme concentration affects initial rate when substrate is abundant versus limiting.
- Q5 (5): Design a temperature experiment that separates thermal acceleration from irreversible denaturation.

Provided evidence or data

Substrate (mM)	Control rate	With inhibitor
0.5	8	3
1	14	6
2	23	12
5	35	27
10	39	36

Submit

- One PDF with a hand- or software-generated rate plot and complete answers.

Grading criteria

Criterion	Pts	Full-credit evidence
Question 1	5	Correct setup, units, and biological interpretation.
Question 2	5	Correct reasoning shown; not just a final answer.
Question 3	5	Mechanism-based explanation uses appropriate evidence.
Question 4	5	Prediction follows from the stated model and conditions.
Question 5	5	Data are represented or evaluated accurately.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.